

PUMA TPC — DLC dispersion, π/K reconstruction, GENFIT vs UKF

Native macOS build: ROOT 6.32 / FairRoot v19 / Geant4 11.2 / GenFit 2.2

What is working on the Mac

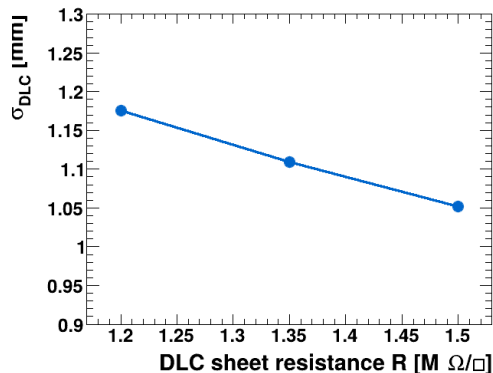
- ✓ Full stack builds natively: FairSoft (ROOT 6.32.08, Geant4 11.2), FairRoot v19, ATTPCROOT, GenFit 2.2
- ✓ **PUMA Geant4 simulation runs** (fixed a Geant4-11.2 production-cut abort)
- ✓ **DLC resistive charge dispersion** from the measured sheet resistance
- ✓ $\pi^+\pi^-$ and K^+K^+ **sims + reconstruction** (pattern recognition)
- ✓ **GENFIT vs UKF comparison**: both fitters run (fixed an Eigen version mismatch that had excluded the UKF); π and K momentum/angle/vertex/charge results
- ✓ **PID study**: dE/dx K vs π separation (Bethe-Bloch) + charge-sign, and the DLC effect

DLC dispersion — the physics

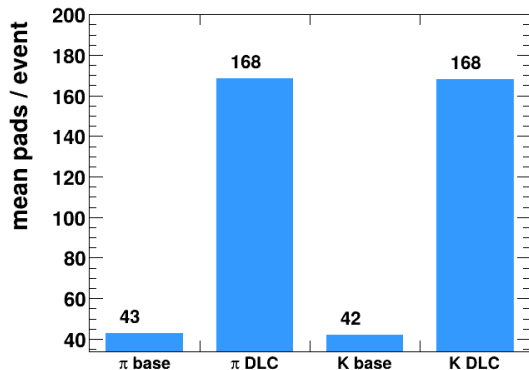
$$\sigma_{\text{DLC}} = \sqrt{2t/(RC)}$$

- R = DLC sheet resistance = **1.2–1.5 M $\Omega/\square$** (measured)
- C = areal capacitance DLC $\leftrightarrow$ pads ($\sim 6 \times 10^{-13}$ F/mm²)
- $t \approx$ shaping time (0.5 μ s)

Replaces the origin package's *assumed* $RC = 100$ ns/mm².
At 4 T gas diffusion is suppressed, so this term dominates.



DLC effect on digitization

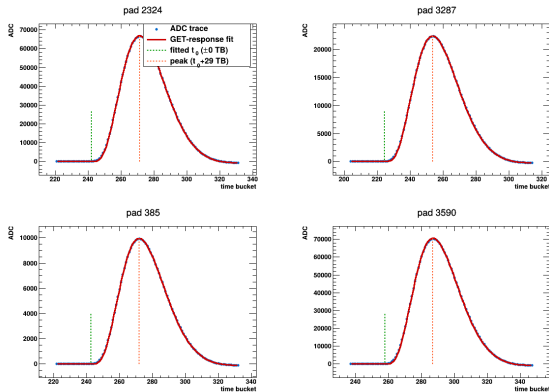


$\sigma_{\text{DLC}} \approx 1.0\text{--}1.2 \text{ mm} \Rightarrow$ charge over $\sim 1\text{--}2$ pads $\Rightarrow$
 $\sim 4 \times$ **pad multiplicity**.

Previously the digitizer used base AtPulse with *no* dispersion.

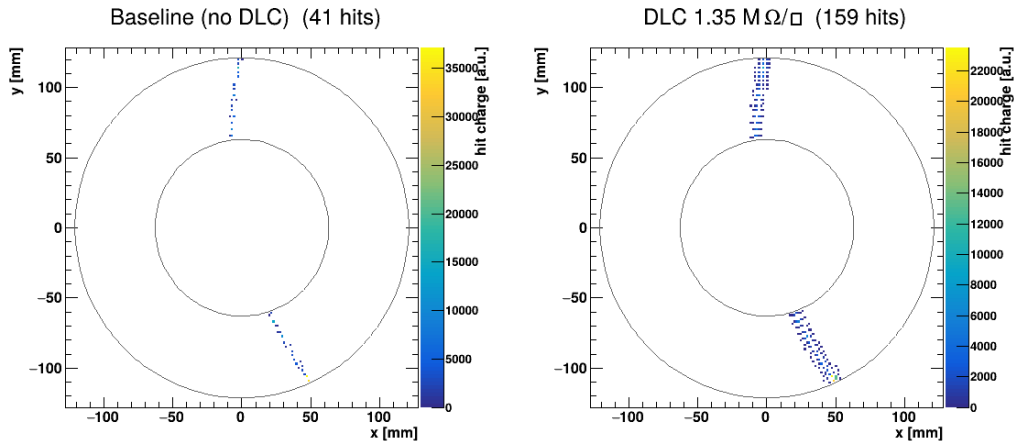
Charge-conserving grid sub-sampling through the annular pad map $\Rightarrow$ sub-pad centroiding.

Pad pulses & the PSA fit



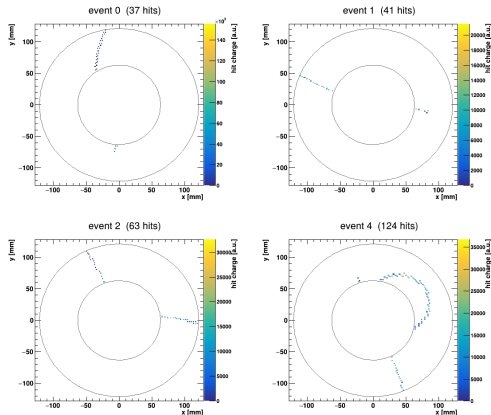
Pad traces (blue) with the **GET electronics-response fit** (red, AtPSAMultiFit: $A e^{-3u} \sin u u^3$, $u = (t - t_0)/\tau$, $\tau = 25$ TB) — overlays the pulse shape **perfectly**. Fitted **impulse time t_0** = **unbiased** drift depth (z =vertex); **response peak** lags t_0 by the ~ 8.6 mm shaping delay. max/mfpeak/mfimpulse all give **equivalent** ~ 16 – 18% momentum.

DLC in a real event ($\pi^+\pi^-$)



Same event, pads on a **charge (ADC) colour scale**: baseline (left, 41 pads) vs DLC (right, 159 pads). DLC spreads each cluster over ~ 2 pads $\rightarrow$ lower per-pad amplitude but sub-pad centroiding.

Reconstructed tracks — pad charge colour scale

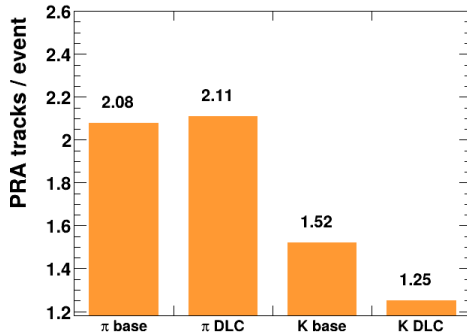


Four events on the annular PUMA pad plane ($R = 62.9\text{--}121.1$ mm). Each hit/pad is drawn on a **charge (ADC) colour scale** (Z-axis palette = deposited charge/pad); tracks appear as the curved chains of fired pads, colour showing where ionization is heaviest. $dE/dx = \Sigma Q_{\text{pad}}/\text{arc}$ is built from exactly these values.

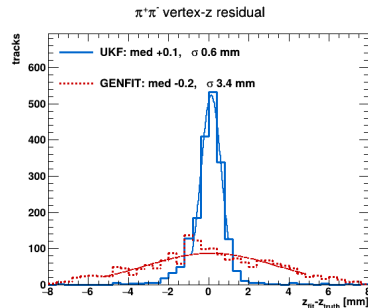
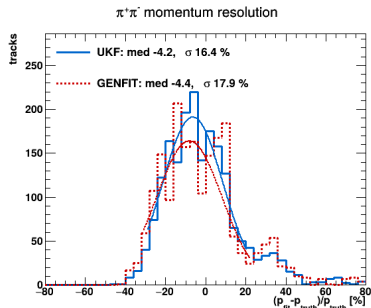
π and K simulations + reconstruction

Sample	pads/evt base→DLC	tracks/evt base→DLC	eff
$\pi^+\pi^-$ 0.4 GeV	42.7→168.3	2.08→2.11	100%
K^+K^+ 0.777 GeV	41.9→168.1	1.52→ 1.25	100%

Same-charge kaons = mirror circles; DLC spreading slightly **merges** them in PRA (1.52→1.25).



UKF vs GENFIT — pion resolution (150 evts)

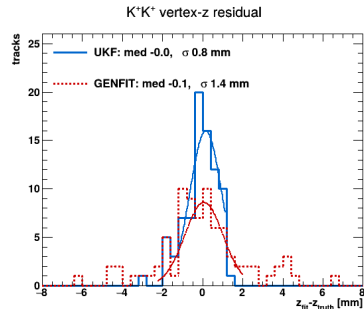
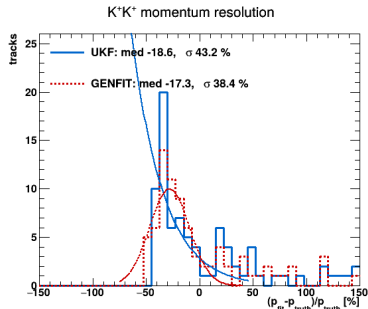


$\pi^+\pi^-$, $|p| = 374.9$ MeV/c, both fitters on *identical* clusters.

	p med/ σ_{IQR}	θ σ	vtx σ_z	charge
UKF	-6.4/ 18.0%	1.9°	0.63 mm	99.2%
GENFIT	-5.9/21.2%	2.6°	1.14 mm	99.2%

Momentum $\sim$ **18–21%**, $\theta \sim 2^\circ$, charge **99%**; **UKF vertex-z 2× tighter**. **Per-ring centroiding** is the key reco choice; the PSA method (max / multi-fit) is equivalent. Robust median/ σ_{IQR} .

UKF vs GENFIT — kaons + per-ring clustering (150 evts)



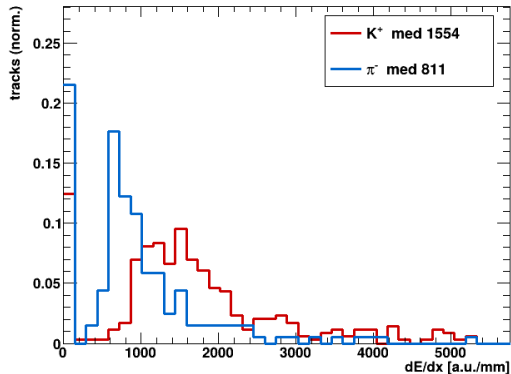
K^+K^+ , $|p| = 600$ MeV/c, per-ring resistive centroiding.

	p med/ σ_{IQR}	θ σ	vtx σ_z	charge
UKF	-18.6/43%	2.4°	0.78 mm	100%
GENFIT	-17.3/ 38%	2.5°	1.42 mm	100%

Per-ring centroiding rescues the kaons: the near-straight guard *rejects* no-curvature tracks (keeps **45%**, 78/174) → momentum **measurable** (~40%) & charge **100%** (was ~74% no-ring). Quality-vs-efficiency: measure the well-curved half cleanly, drop the rest.

Particle ID — dE/dx (K vs π)

PUMA dE/dx PID (base): separation = 0.75



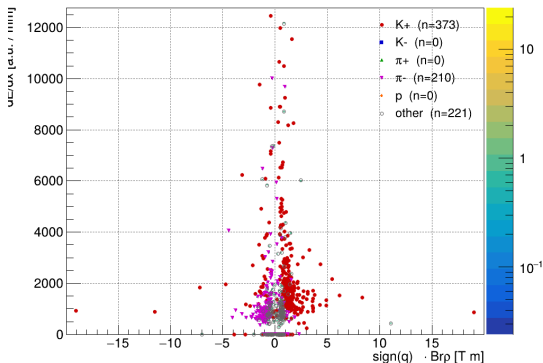
Full channel ${}^3\text{He} + \bar{p} \rightarrow K^+ K^+ \pi^- + \text{hex}$ (both species/event, truth-labelled); $\text{dE/dx} = \Sigma Q_{\text{hit}}/\text{arc}$:

	K med	π med	K/ π	sep
base	1554	811	$1.9\times$	0.75σ
DLC	327	172	$1.9\times$	0.72σ

K deposits $\sim 1.9\times$ the π dE/dx (Bethe-Bloch), $\sim 0.75\sigma$ separation. **DLC** rescales dE/dx but leaves separation unchanged. Charge-sign (prev. slide) completes the PID.

PID map — dE/dx vs $B\rho$

PUMA PID - 804 tracks - $B = 4.0$ T



The classic 2D PID map: dE/dx vs $\text{sign}(q) \cdot B\rho$, with $B\rho = BR/1000$ [T m] from the fitted circle radius R in 4 T.

Two handles combine: **charge sign** splits the plane left/right (π^- vs K^+); dE/dx (vertical) adds the mass ordering ($K^+ \sim 1.9\times$ higher).

Truth-labelled, full $K^+K^+\pi^-$ + hex channel: K^+ (n = 373), π^- (n = 210). The plot a real analysis cuts on; K^+K^+ merging populates the residual overlap.

How the fitter comparison was unblocked

The UKF was *never compiled* — root cause was an **Eigen version mismatch**:

- brew installed **Eigen 5.0.1**; ATTPCROOT does `find_package(Eigen3 3.3)`. Eigen 5.x is declared incompatible with a 3.3 request → `Eigen3::Eigen` target never created → the `if(TARGET Eigen3::Eigen)` block (AtFitterUKF, OpenKF) skipped → symbol absent from lib & dictionary.
- **Fix:** installed **Eigen 3.4.0**, prioritised over brew's. Rebuilt → UKF compiles → interpreted macro runs both fitters (as on Linux). Baked into `env.sh`.

Summary & reproduce

Native macOS PUMA chain sim $\rightarrow$ digi(DLC) $\rightarrow$ PSA $\rightarrow$ PRA $\rightarrow$ UKF/GENFIT works.

DLC 1.2–1.5 M Ω /□ $\rightarrow$ $\sigma \approx 1.0$ –1.2 mm $\rightarrow \sim 4\times$ pads. Both fitters run, UKF $\approx$ GENFIT. π : p -res ~ 18 –21%, $\theta \sim 2^\circ$, charge 99%. K: per-ring clustering $\rightarrow p$ measurable ($\sim 40\%$), charge 100% (45% eff). Key reco knob = ring centroiding (PSA max/multi-fit equivalent).

Reproduce (branch puma-genfit-pion-mac)

```
source build/config.sh
root -b -q 'PUMA_test8_sim.C(150,0.4)'      # pion sim (test10 -> kaon)
root -b -q 'run_digi_ukf_genfit_test8.C(150,8.,false,"max",\
    "pi",8,16,256,0.,true,true,false,20)'  # max PSA + ring clustering
root -b -q 'compare_ukf_genfit_test8.C("pi")' # resolution numbers
root -b -q 'resolution_plots.C("pi")'      # dp/theta/vz figs
```

Still to confirm: C (DLC areal capacitance) — the only free parameter in σ_{DLC} .